The New Scientific Method

Will Handley
<wh260@cam.ac.uk>

Royal Society University Research Fellow
Institute of Astronomy, University of Cambridge
Kavli Institute for Cosmology, Cambridge
Gonville & Caius College
willhandley.co.uk/talks

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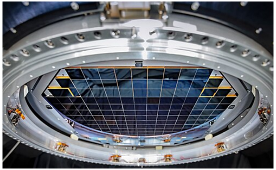
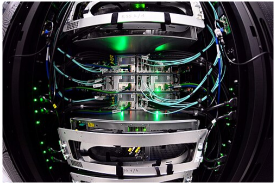
The CCD Analogy

A change in how we capture data → a change in how we analyse it



1980s–2000s: Observation

- **Photographic Plates → CCDs**
- Digital sky surveys (SDSS → Rubin/LSST)
- Quantitative, survey-scale astronomy

2020s: Analysis

- **CPUs → GPUs**
- **Manual Coding → LLM-Accelerated**
- Real-time inference

Outline: Two revolutions

Distinct shifts, often conflated

1. GPUs for analysis

- GPU hardware accelerates *classical* statistical methods
- Gravitational Waves, Dark Energy, Supernovae, Radio
- The method transfers across the department

2. LLMs for development

- How we built this in months, not years — and we're just getting started
- Verification, legacy code, and new workflows
- Expanding the scope of what a small group can tackle

3. Live Demo & Closing

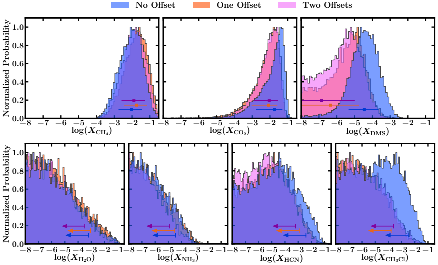
Loop running in front of an audience: the most dangerous game.

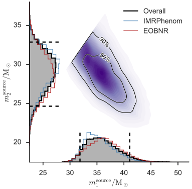
What I mean by "classical inference"

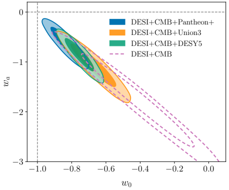
Triangle plots, error bars, and Bayesian model comparison

- Algorithms that converge to the true posterior: MCMC, Nested Sampling, HMC
- **JWST**: K2-18b atmospheric biosignatures [2309.05566]
- **LIGO**: gravitational wave detection [1602.03837]
- **DESI & DES**: dark energy and cosmic shear [2503.14738] [2601.14559]

- The instruments exist. The statistical methods exist.
- The bottleneck is compute: running them fast enough at scale.







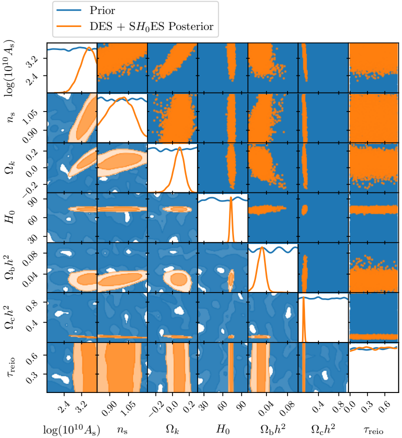
Bayesian model comparison at scale

`unimpeded`: 248 dataset–model combinations via nested sampling

Dily Ong
PhD Student

- Classical inference "at scale" — without GPUs
- 248 pairwise dataset–model combinations: full Bayesian evidences and tension statistics
- Tens of millions of CPU-hours via DiRAC grants
- Planck, DESI, DES, Pantheon, ACT, SPT, SH0ES, . . .

[2511.05470] Nested sampling database
[2511.04661] Model comparison & tension grid
[2511.10631] Bayesian evidence for evolving DE



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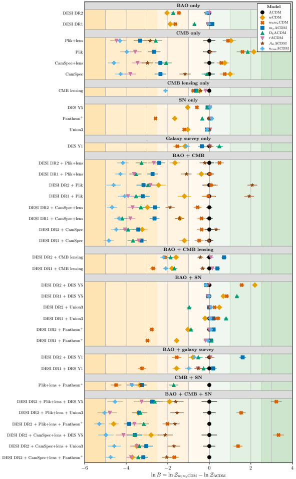

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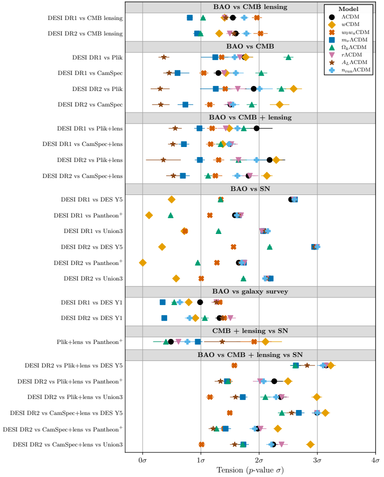


David Yallup

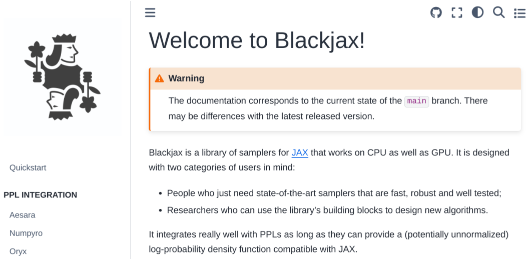
Postdoc

- These analyses work, but they're slow. If we can speed them up, new science becomes possible:

  Robustness Marginalise over 100+ nuisance parameters rather than fixing them.

  Discovery at scale Definitive Bayesian evidence for new physics across grids of models and datasets.

  Low latency Inference fast enough for triggers, scheduling, and real-time follow-up.



- Nested Sampling on a GPU [2601.23252]
- Alongside MCMC, HMC, SMC [2402.10797]
- In Google's premier GPU language
- *No gradients required.*

GW170817: the multi-messenger event

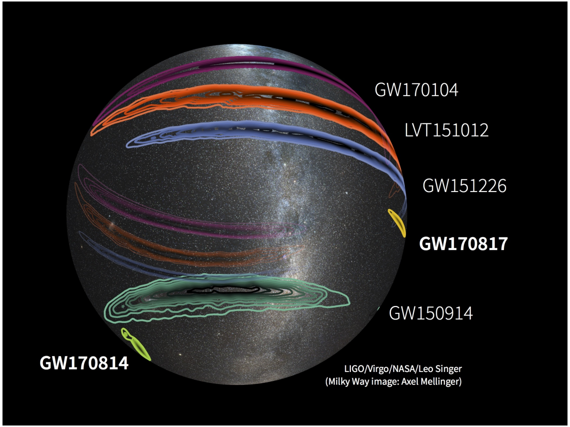
Example 1: Real-time gravitational wave parameter estimation

Sky localisation and EM follow-up [1710.05833]

- Binary neutron star merger detected by LIGO/Virgo
- EM follow-up required fast sky localisation
- Triggered >70 observatories worldwide

The bottleneck

- Full parameter estimation (`bilby`): **60 CPUs, 2 hours**
- Too slow for prompt follow-up of short-lived transients



LIGO/Virgo/NASA/Leo Singer

30 seconds on one GPU

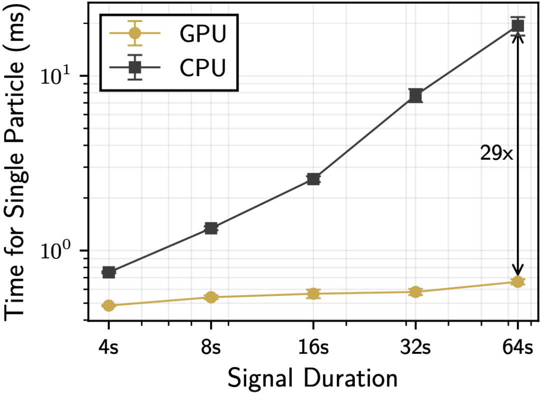
Example 1: Real-time gravitational wave parameter estimation

GPU nested sampling [2509.04336]

- **1 GH200 GPU** for ~**30 seconds**
- 4 s of real LIGO data (GW150914), full 15-D parameter space
- Standard CPU: 60 CPUs × 2 hours

What near-real-time means

- Close enough to scan the time stream continuously
- EM alerts while the event is still physically interesting
- No more cheap template banks — full waveform models find more candidates



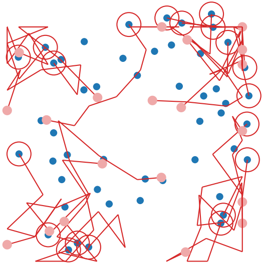
Where does the speed come from?

Example 1: Real-time gravitational wave parameter estimation

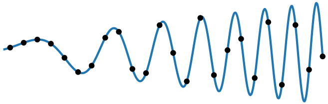
The Bottleneck [2509.04336] [2509.24949]

- Waveform generation (physics) is *not* the bottleneck — memory bandwidth for extrinsic parameters is.
- Dual-axis parallelisation: over sampling particles *and* frequencies in the likelihood.
- Most codes use a tiny fraction of GPU — this opens a different scaling relation for next-gen detectors.

Inter-likelihood: parallelise over particles



Intra-likelihood: parallelise over frequencies



[github:mrosep/talks]

Is dark energy evolving?

Example 2: Dark energy with DESI and DES

Dily Ong

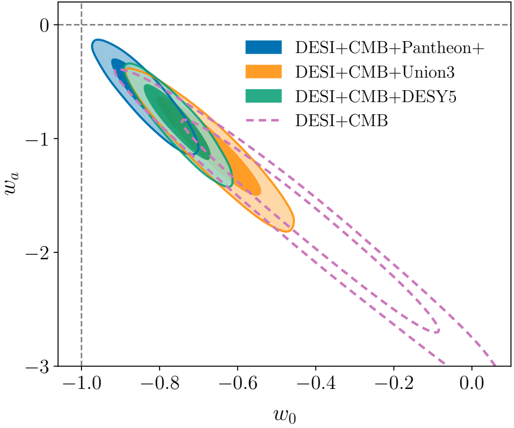
PhD Student

The DESI claim [2503.14738]

- DESI DR2: $w_0 w_a$CDM preferred over ΛCDM at 3.1σ (BAO+CMB), up to 4.2σ with SNe.
- Frequentist $\Delta\chi^2$ with only 7 BAO data points.

Bayesian reanalysis [2511.10631]

- BAO+CMB alone: ΛCDM modestly favoured ($\ln B = -0.57 \pm 0.26$).
- Only reaches $\sim 3\sigma$ *with* DES-Y5 SNe — driven by inter-dataset tension, not DE evolution.



Beyond $w_0 w_a$: flexible dark energy on GPU

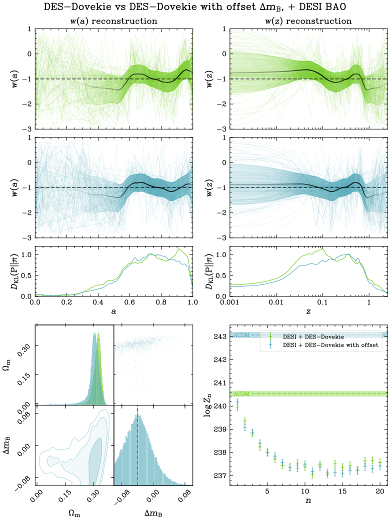
Example 2: Dark energy with DESI and DES

Adam Ormondroyd
PhD Student

Flexknot $w(a)$ reconstruction [2503.08658] [2509.13220]

- $w_0 w_a$ is too restrictive — flexknots let the data speak.
- W-shaped structure in $w(a)$; DESI and DES-5Y are in tension.
- A magnitude offset Δm_B between low-z and DES SNe beats flexknot DE in Bayesian evidence.

20 model complexities × multiple datasets — only feasible because the pipeline is in JAX: **days → minutes**.



Lawrence Berry

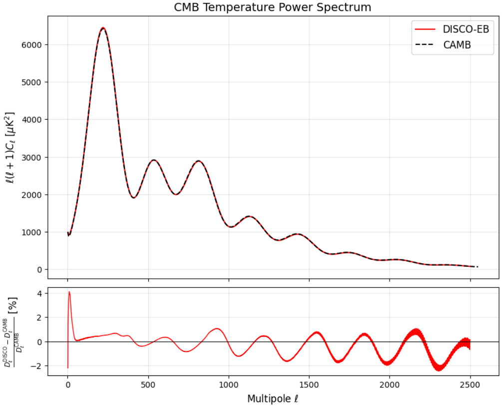
PhD Student

The missing component

- BAO and SNe are in JAX. The CMB is not — CAMB/CLASS are legacy Fortran/C.
- To fold in Planck/ACT properly, we need a GPU-native Boltzmann solver.

DISCO: GPU-native Boltzmann solver

- Full Boltzmann solver in JAX, GPU-native, auto-differentiable.
- Sub-percent agreement with CAMB (0.7% mean residual).
- $\sim$5 s per model on H100 and falling; saturates well on a GPU.



GPU-accelerated supernova photometry

Example 3: Supernovae as standard candles

Sam Leeney

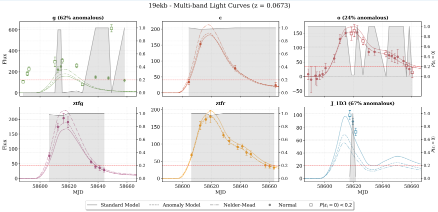
PhD Student

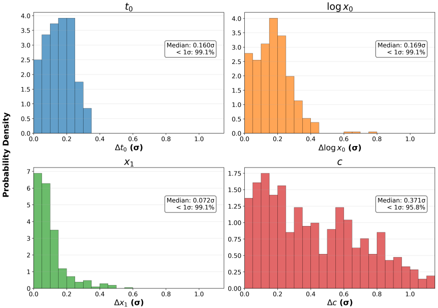
jax-bandflux: SN photometry on GPU [2504.08081]

- JAX SALT3 bandflux integration $\Rightarrow \sim$**100**$\times$ **speedup**, fully differentiable.

Bayesian anomaly detection [2509.13394]

- Each data point gets a latent anomaly probability, fitted jointly with SALT3 parameters.
- Automates outlier flagging, filter rejection, *and* data preservation within partially contaminated bands.
- 98% of parameters match manual analysis — but fully automated for LSST.





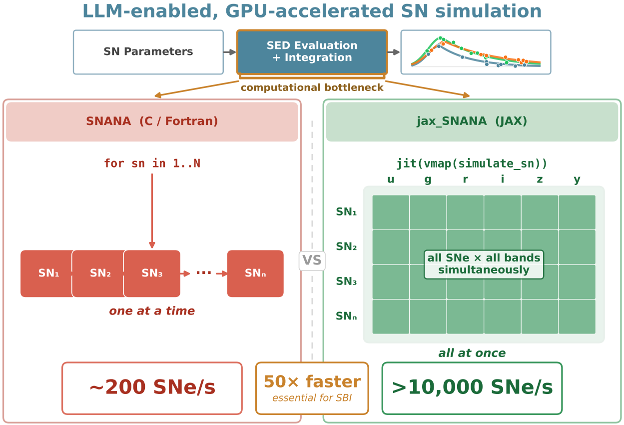
Matt Grayling
Postdoc

Simulations at scale

Example 3: Supernovae as standard candles

JAX-SNANA: 10,000 sims/sec on A100

- Full SN light-curve simulation pipeline rewritten in JAX.
- $\sim$200 sims/sec (CPU) $\rightarrow$ $>$**10,000/sec** (GPU): 50$\times$ speedup.
- Validated against DES Y5 to <1 millimag. Essential for SBI at LSST scale.



Toby Lovick

PhD Student

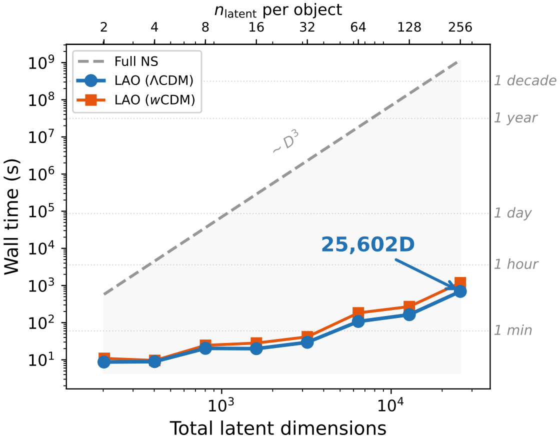
Laplace Approximation Optimisation

- SALT2 standardises with 2 parameters per SN. A physical model could use 100s of measurements per object.
- **25,602-D** likelihood in **11 minutes**. NS: $D^3 \to$ a decade of H200 time.

Autodiff for marginalisation [2406.08703]

- Optimise to mode, auto-diff the Hessian, marginalise exactly. Parallelised across objects on GPU.
- No manual derivations — auto-diffs the forward model. Change the physics, re-run.

(a) Computational cost



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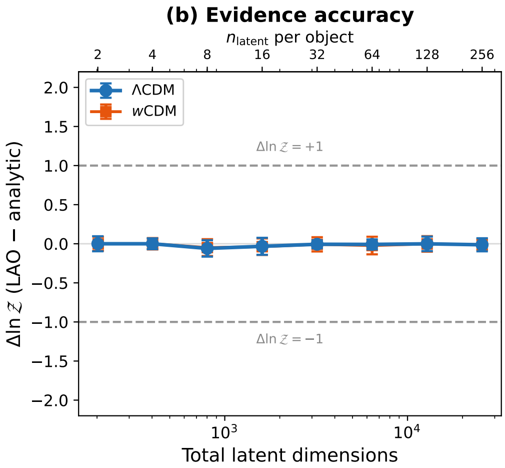


21cm Radio Astronomy

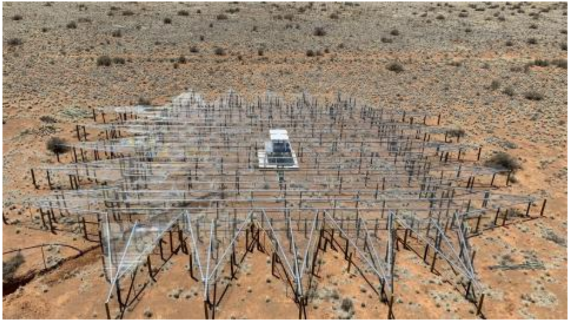
Sublinear scaling enables new science [2210.07409]

REACH GPU Pipeline

- $\mathcal{O}(N^{0.13})$ scaling for likelihood evaluations.
- $\sim$0.15 ms per likelihood call for a full year of data ($\sim 10^5$ observations).

Bayesian Anomaly Detection

- Full year of real REACH data: $\sim$1 hour on one GPU.
- Anomalies correlate with LST, time of night, season — fully automatic, no manual flagging.



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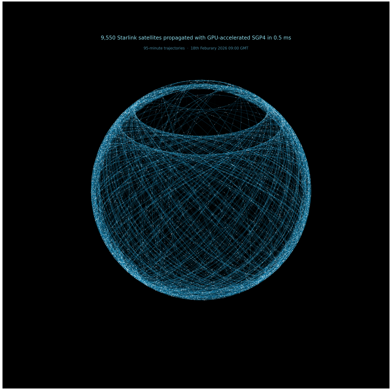
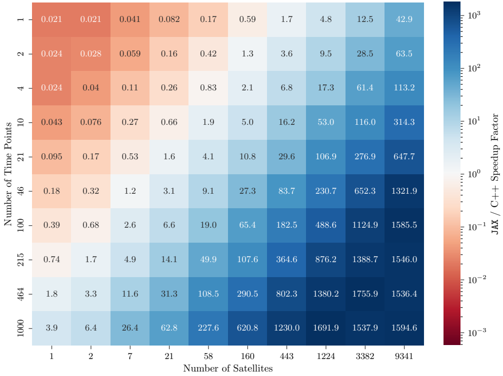
Satellites [GLITTER]

JAX-SGP4: GPU-accelerated orbit propagation

Charlotte Priestley

PhD Student



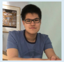


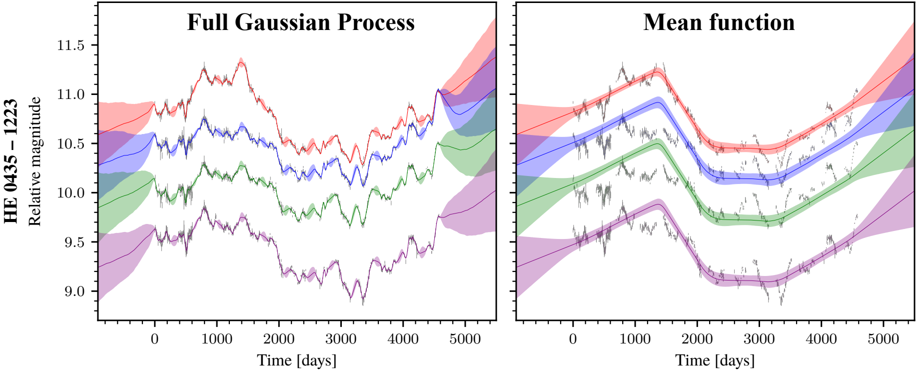
JAX-SGP4: Propagating 9,550 Starlink satellites in 0.5 ms. 1550× speedup on A100.

Quasar Time Delays

Robust inference via flexible Gaussian Processes

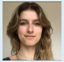
Namu Kroupa
PhD Student

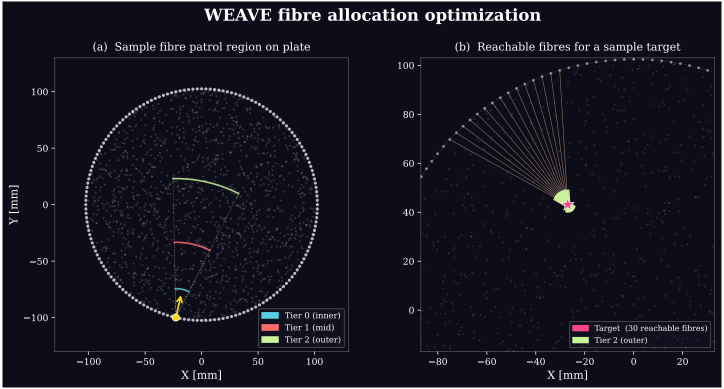




Marginalising over GP kernels with nested sampling makes time-delay inference robust to modelling assumptions. *(with Alex Drane: 150× GPU speedup.)*

Farah Wallauer
Part III Student





WEAVE: 168 retractors, ∼1000 fibres — collision-free assignment via JAX on GPU. 60× speedup over C++ simulated annealing.

Galaxies

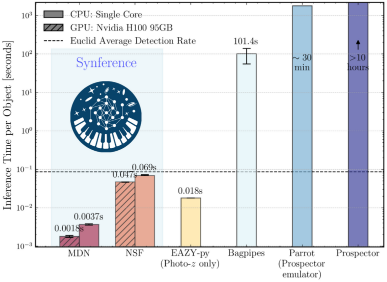
GPU-accelerated SED generation

Rose Luo
Part III Student

Chris Lovell
Postdoc

SYNTHESIZER

synthesizer-project.github.io
Lovell et al. 2025, Roper et al. 2025
Harvey et al. 2026



Synthesizer: Forward modelling galaxy emission. **Synference**: SBI for SED fitting — orders of magnitude faster, full posteriors, applications for Euclid.

Part 1: GPUs for science ✓

Part 2: LLMs for scientists

How we built all of this in months, not years

A brief history

How we got here, fast

- 2019 Tabnine — deep learning code completion
- Jun 2021 GitHub Copilot — AI pair programming
- Nov 2022 ChatGPT (GPT-3.5) — "the moment"
- Mar 2023 GPT-4 — qualitative leap
- Dec 2023 Gemini 1.0 — Google enters the race
- Jun 2024 Claude 3.5 Sonnet — the coding model
- Sep 2024 OpenAI o1 — reasoning models
- Oct 2024 Cursor — agentic IDE
- Jan 2025 DeepSeek R1 — open-source reasoning
- May 2025 Claude Code — agentic coding CLI
- Aug 2025 GPT-5 — unified model
- Nov 2025 Gemini 3, Opus 4.5
- Dec 2025 GPT-5.2
- 2025– Qwen, Kimi, GPT-OSS — open source catches up
- Feb 2026 Opus 4.6 — *this talk*

Three ways AI meets science

Only one matches what science actually is

Foundation Models

- “Train a model on all astronomical data.”
- Expensive, opaque, unclear payoff.
- Physics is about *compression*, not prediction.

Autonomous Agents

- “Click a button, generate a paper.”
- Produces “science-like” content without understanding.
- Science is a human culture of critique — not a pipeline.

Augmented Humans

- Humans accelerated, not replaced.
- Humans define the physics and verification; AI handles the syntax.
- **Our approach.**

“Silicon Valley thinks a scientist is something that takes in coffee and produces fusion reactors. It’s not.”

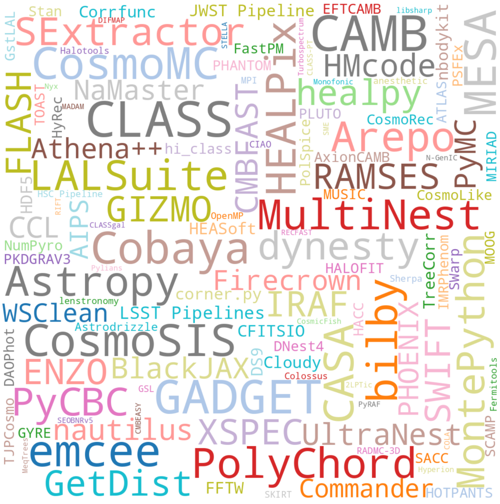
The code debt problem

Why PhD students spend years operating frameworks instead of doing physics

- Our fields are riven with **code debt**.
- Students end up *operating* CAMB, CLASS, Cobaya — not implementing their own ideas.
- We mistake "ability to debug legacy Fortran" for "scientific aptitude."

The shift

LLMs are autocomplete on steroids — our route out of the spiral. We move from **maintaining code** to **specifying physics**.



The mechanism: context engineering

From "prompting" to "orchestration"

2025: Prompt engineering

- *"Write me a Python loop."*
- Limited by user memory and syntax knowledge.
- Useful for snippets, bad for systems.

2026: Context engineering

- *"Here is the CAMB repo. Find the neutrino integration logic."*
- LLMs acting on the **full context** of the project.
- The bottleneck is no longer generating code, but **evaluating** it.

LLMs as interactive documentation

Navigating and excavating legacy codebases

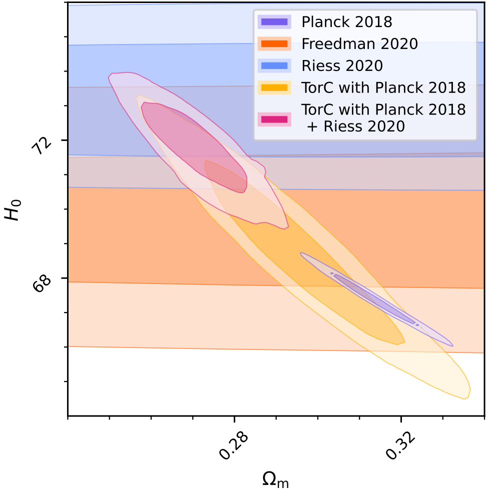
Sinah Legner
PhD Student

Sinah: Torsion Condensation in CAMB [2507.09228]

- LLM traces dependencies across CAMB's Fortran codebase.
- Found N_{eff} double-counting bug.
- Diagnosed missing massive neutrino evolution.

Will: CosmoSIS → Cobaya

- Porting DES Y3/Y6 likelihoods between frameworks.
- Adversarial review: one LLM writes, another critiques.



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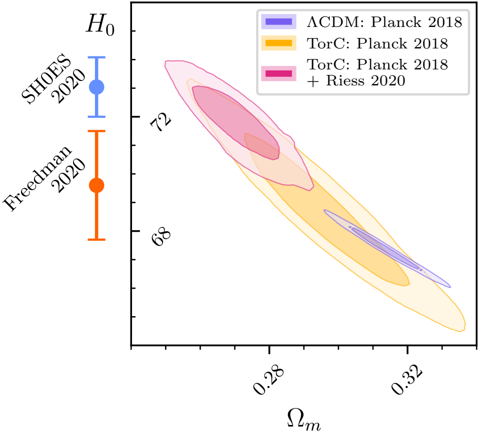


Deep surgery on C code: CLASS & curvature

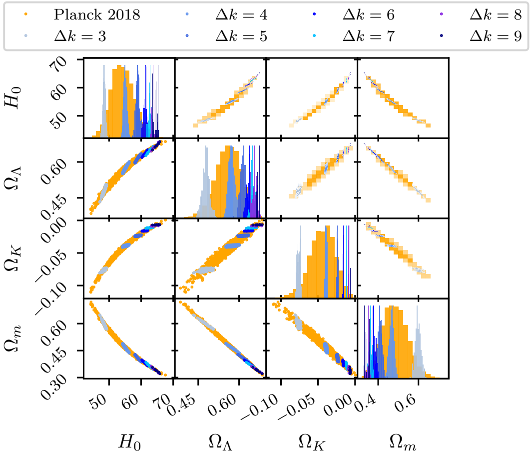
Solving boundary value problems with AI assistance [2407.18225] [2509.10379]

Discrete k-quantisation in CLASS

- Enforcing discrete k-modes for closed universe topology.
- Matching boundary conditions at the Big Bang *and* Future Conformal Boundary.

LLM role: debugging & code navigation

- Claude Code for testing, catching numerical errors, modifying C code.
- Core physics from reading the papers — LLM accelerates the engineering, not the science.



Deep surgery on C code: CLASS & curvature

Wei Ning Deng
PhD Student

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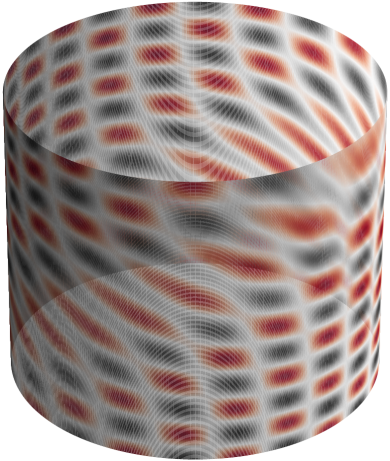
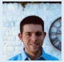


Development time collapsed

The work didn't disappear — it moved to verification

James Alvey
KICC Fellow



Natalie Hogg
Postdoc

Implementation `emcee` in BlackJAX: 2 months (manual) → **15 minutes** (LLM).

Optimisation Natalie Hogg: covariance matrix 24 h → **1–2 min**. Vectorisation, JIT, quasi-Monte Carlo.

Portability Sam Leeney: REACH anomaly detection → supernova cosmology in hours. [2504.08081]

New science GW emulator with James Alvey: nothing → working package in **8 hours**.

The insight

I mistook *mental athleticism* — typing speed, syntax memory, algebraic calculation — for scientific strength.

With AI, **scientific principles** and **creative drive** become the differentiators.

"We didn't feel tired — we were spending all our time reasoning about physics."

Verification is the new lab bench

LLMs make wrong code easy. Testing makes it detectable.

Reference outputs

Match legacy code to floating-point precision.
Regression to known-good results.

Simulation-based calibration

KS-tests on samples (SBC).
Statistical validation of the full pipeline.

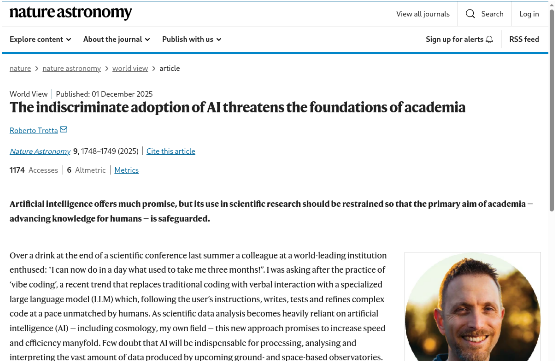
Independent testing

One person implements, another writes tests.
Adversarial review: one LLM writes, another critiques.

"We are no longer limited by how fast we write code — we are limited by how rigorously we verify it."

Two serious arguments

Both from February 2026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World View | Published: 01 December 2025

The indiscriminate adoption of AI threatens the foundations of academia

Roberto Trotta

Nature Astronomy 9, 1748–1749 (2025) | Cite this article

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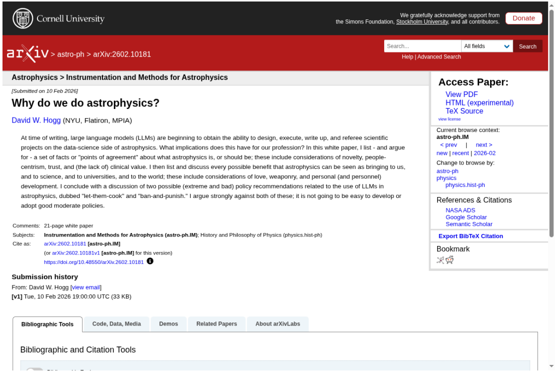
Artificial intelligence offers much promise, but its use in scientific research should be restrained so that the primary aim of academia — advancing knowledge for humans — is safeguarded.

Over a drink at the end of a scientific conference last summer a colleague at a world-leading institution enthused: "I can now do in a day what used to take me three months!". I was asking after the practice of 'vibe coding', a recent trend that replaces traditional coding with verbal interaction with a specialized large language model (LLM) which, following the user's instructions, writes, tests and refines complex code at a pace unmatched by humans. As scientific data analysis becomes heavily reliant on artificial intelligence (AI) — including cosmology, my own field — this new approach promises to increase speed and efficiency manyfold. Few doubt that AI will be indispensable for processing, analysing and interpreting the vast amount of data produced by upcoming ground- and space-based observatories.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We gratefully acknowledge support from the Simons Foundation, Stockholm University, and all contributors. Donate

arXiv > astro-ph > arXiv:2602.10181

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Astrophysics > Instrumentation and Methods for Astrophysics

[Submitted on 10 Feb 2026]

Why do we do astrophysics?

David W. Hogg (NYU, Flatiron, MPIA)

At time of writing, large language models (LLMs) are beginning to obtain the ability to design, execute, write up, and referee scientific projects on the data-science side of astrophysics. What implications does this have for our profession? In this white paper, I list - and argue for - a set of facts or "points of agreement" about what astrophysics is, or should be; these include considerations of novelty, people-centrism, trust, and (the lack of) clinical value. I then list and discuss every possible benefit that astrophysics can be seen as bringing to us, and to science, and to universities, and to the world; these include considerations of love, weaponry, and personal (and personnel) development. I conclude with a discussion of two possible (extreme and bad) policy recommendations related to the use of LLMs in astrophysics, dubbed "let-them-cook" and "ban-and-punish." I argue strongly against both of these; it is not going to be easy to develop or adopt good moderate policies.

Comments: 21-page white paper

Subjects: **Instrumentation and Methods for Astrophysics (astro-ph.IM)**; History and Philosophy of Physics (physics.hist-ph)

Cite as: arXiv:2602.10181 **[astro-ph.IM]**

(or arXiv:2602.10181v1 **[astro-ph.IM]** for this version)

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Two serious arguments

Both from February 2026

Trotta [2602.10165]

- LLM reliance causes measurable cognitive decline (Kosmyna et al. 2025).
- Scientific coding requires understanding the models — outsourcing this undermines science.
- Students risk becoming "little more than prompt engineers."

Hogg [2602.10181]

- Students are ends, not means — "better than a grad student" is a dangerous framing.
- LLMs cannot take responsibility or be authors.
- Neither "let them cook" nor "ban and punish" works — no clear middle way.

My response

I agree with both. Use LLMs for overhead, not for the science. Verify rigorously.

I wrote this talk without typing

What I used

- **Voice**: Otter.ai transcription → notes
- **Loop runner**: Claude Code (Opus 4.6)
- **Reviewers**: Gemini 3 and GPT-5.2 as independent critics
- **Knowledge base**: 1 year of 450 transcripts, whiteboard photos, emails, LaTeX of papers and grants — in a context management system
- **Email**: 14 personalised figure requests, sent and tracked
- **Web**: Playwright, arXiv, Wikimedia Commons
- **Drafting**: Gemini (1M context) for slides from source material

What I did

- Talked about the science
- Chose the figures
- Edited the emails
- Said "no" a lot
- Verified the physics

What I didn't do

Type. Read Fortran. Manually search 450 transcripts. Write 14 emails from scratch. Make a single LaTeX file by hand.

Live Demo

The generate → test → iterate loop, not one-shot codegen

[Claude Code Demo]

Conclusions

GPUs $\neq$ AI

- Classical statistics on GPU is competitive with ML — and interpretable.
- GW PE: 2 h → 30 s. Dark energy: days → minutes. Supernovae: 50×. The method transfers across astronomy.

LLMs $\neq$ science

- LLMs don't replace scientists — they make code and text virtually free.
- The bottleneck shifts from *writing* to *verifying*. Verification is a higher-order scientific skill.

Seize the opportunity

Those who combine rigorous inference with LLM-accelerated development will lead the next decade of astronomy. Capture content. Engineer context. Stay the scientist.

If producing code and text were **virtually free**, *what would you do differently as a scientist?*

handley-lab.co.uk [github:handley-lab]